Two-Stage False Data Injection Attack Detection in EV Charging Networks Using NARX Neural Network with Isolation Forest and CUSUM Temporal Confirmation

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***Abstract*—False Data Injection (FDI) attacks represent a critical threat to Electric Vehicle (EV) charging infrastructure, where adversaries manipulate charging management signals to disrupt session parameters, causing financial and safety harm. While neural network-based energy estimation using Nonlinear AutoRegressive networks with eXogenous inputs (NARX) has demonstrated promising results in detecting such attacks via Error-of-Estimation (EoE) analysis, existing detection layers suffer from fundamental limitations: static global threshold methods such as IQR are blind to subtle, short-duration attacks, their parameters are set heuristically, and session-boundary transitions pollute global anomaly bounds. In this paper, we retain the NARX estimator from the state-of-the-art baseline and replace its single-stage anomaly detector with a principled two-stage hybrid: an Isolation Forest (IF) performs a high-sensitivity sweep that guarantees zero missed attacks, and a Cumulative Sum (CUSUM) control chart then filters candidate anomalies by confirming only *sustained* EoE drift patterns consistent with the physical signature of an FDI attack. CUSUM is motivated by the attack physics directly: a manipulated `minutesAvailable` parameter causes EV charging to halt for multiple consecutive timestamps, creating a prolonged directional shift in EoE that CUSUM is specifically designed to detect, unlike point-wise outlier methods. Evaluated on the Caltech ACN-Data dataset (December 2020–January 2021), the proposed IF+CUSUM pipeline achieves a perfect recall of 100% (zero missed attacks) with precision of 61.8% and an F1 score of 0.764, compared to F1 = 0.682 and recall = 57.9% for the Global IQR baseline at the same operating point. An ablation study across attack intensities $\vartheta \in \{1, 5, 10, 20, 30, 60\}$ minutes further demonstrates that the proposed method detects subtle attacks ($\vartheta < 10$ min) that are entirely invisible to static percentile thresholds.**

***Index Terms*—deep learning, electric vehicles, EV charging, false data injection, NARX neural network, Isolation Forest, CUSUM, anomaly detection, cybersecurity, smart grid**

I. Introduction

The rapid proliferation of EV charging infrastructure has transformed urban mobility, yet it has simultaneously expanded the attack surface of the smart grid. EV Supply Equipment (EVSE) and Charging Management Systems (CMS) exchange sensitive session parameters—such as pilot signal, user identity, and availability duration—over open network protocols, most commonly the Open Charge Point Protocol (OCPP). An adversary who intercepts or spoofs these messages can inject false data that alters the energy delivered to a vehicle, leading to battery over-charge, under-delivery, or billing fraud [1].

FDI attacks are particularly dangerous in the EV context because they are *stealthy*: a small perturbation of the scheduled charging time (e.g., reducing `minutesAvailable` by a few minutes) may produce a deviation in delivered energy too small for a human operator to notice, yet sufficient to degrade vehicle range or billing integrity across thousands of sessions. Conventional rule-based anomaly detectors rely on deterministic thresholds derived from historical statistics, which cannot adapt to the diverse, multi-modal EoE distributions produced by heterogeneous charging sessions and fleet compositions [12].

Machine learning approaches have been applied to FDI detection in smart grids, including LSTM-based classifiers [7], CNN-based spatial pattern recognition, and residual-based model checking [4]. Most recently, Basumatary et al. [1] proposed using a NARX neural network as an energy estimator and applying Global IQR statistics on the resulting EoE to detect attack-induced deviations. Their system achieves an impressive 97.5% recall under a fixed attack intensity; however, they explicitly acknowledge three limitations that motivate the present work:

1) The method *"faces challenges in identifying attacks that sustain only for a very short duration or involve very subtle deviations in EoE"* [1]—a gap we term the *subtle attack blindspot*.
2) The IQR multiplier $\beta = 5$ and sliding window $q = 5$ are set heuristically with no principled selection methodology.
3) Computing IQR globally across all concatenated sessions means that cold-start spikes at session boundaries contaminate the anomaly bounds—the paper itself notes these are detected spuriously.

We address all three gaps by replacing the anomaly detection layer with a two-stage hybrid detector that retains the proven NARX estimator unchanged. **Stage 1** employs an Isolation Forest [14] trained on clean (non-attack) EoE to flag all candidate anomalies with perfect recall. **Stage 2** applies a CUSUM control chart [13] to confirm only *sustained* EoE drift patterns, eliminating momentary spikes that cause the Isolation Forest's false positives. The logical AND combination of both stages produces a detector that, for the first time in this

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problem domain, simultaneously achieves zero missed attacks and a principled, physics-motivated false positive suppression mechanism.

The main contributions of this paper are:

- A two-stage IF + CUSUM anomaly detection layer that replaces static percentile thresholds with a learned, distribution-agnostic first stage and a physically motivated temporal confirmation second stage.
- A formal connection between the FDI attack physics (energy-delivery halt over multiple timestamps) and the CUSUM detection mechanism (persistent upward shift), providing a theoretically grounded justification for the design choice.
- Principled parameter tuning for CUSUM drift allowance k and alarm threshold h on normal training EoE, with a recall constraint to prevent overly conservative configurations.
- A systematic ablation study across attack intensity $\vartheta \in \{1, 5, 10, 20, 30, 60\}$ minutes demonstrating robustness to subtle attacks that existing methods cannot detect.

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces the technical preliminaries. Section IV describes the proposed detection scheme. Section V presents experimental results. Section VI discusses findings, limitations, and implications. Section VII concludes the paper.

II. Related Work

A. Model-Based FDI Detection

Model-based methods construct an analytical residual between a state estimate and the actual measurement; an alarm is raised when this residual exceeds a bound derived from system noise statistics [4]. While theoretically principled for linear, fully observable systems, these methods degrade significantly when the target system (heterogeneous EV charging with non-stationary load profiles) violates the linearity or observability assumptions, limiting their applicability to large-scale charging networks.

B. Classifier-Based ML Approaches

Deep learning classifiers—including LSTM [7], bidirectional GRU, and CNN variants—have been applied to FDI detection by framing the problem as binary sequence classification. However, these supervised approaches require balanced, labelled attack datasets that are difficult to collect in operational charging networks. Moreover, the original paper's comparison [1] reports that an LSTM classifier achieved 0% precision and recall on the ACN-Data task, suggesting that direct classification on raw charging signals is inadequate without an intermediate residual representation.

C. Regression-Based Estimation

Basumatary et al. [1] introduced the paradigm of using a NARX neural network as an energy estimator whose prediction residual (EoE) forms the detection signal. This separation of concerns —train a regression model on legitimate data, then monitor its residuals—is conceptually clean and avoids the need for labelled attack training samples. Our work adopts this paradigm wholesale and focuses exclusively on improving the anomaly detection layer applied to the EoE signal.

D. CUSUM in Fault and Intrusion Detection

The CUSUM chart, introduced by Page [13], is a classical sequential hypothesis test for detecting persistent shifts in a monitored signal. It has been applied to fault detection in power systems [12], sensor spoofing detection in cyber-physical systems [4], and network intrusion detection. Its key advantage over point-wise threshold methods is sensitivity to directional drift accumulation: a small but persistent offset will grow S(t) monotonically until it crosses the alarm threshold, regardless of whether any single observation is individually anomalous.

E. Isolation Forest for Anomaly Detection

Isolation Forest [14] constructs an ensemble of random trees and scores anomalies by their isolation depth—outliers require fewer partitioning cuts and thus receive lower (more negative) anomaly scores. Unlike density- or distance-based methods, IF makes no distributional assumption and scales linearly with sample size. In the FDI context, where EoE distributions are multi-modal and session-dependent, this distribution-agnostic property is a significant advantage over IQR-based bounds that implicitly assume a unimodal, symmetric error distribution.

III. Preliminaries

A. NARX Neural Network

A Nonlinear AutoRegressive network with eXogenous inputs (NARX) models the current output as a nonlinear function of past exogenous inputs and past outputs [15]:

$$y(t) = F\big(u(t-1), \ldots, u(t-m_x),\, y(t-1), \ldots, y(t-m_y)\big), \tag{1}$$

where $u(t) \in \mathbb{R}^{n_u}$ is the exogenous input vector at time t, $y(t) \in \mathbb{R}$ is the scalar output, m_x and m_y are the exogenous and feedback delay orders, respectively, and F is approximated by a multilayer perceptron with sigmoid activations.

Following [1], we use $m_x = m_y = 2$, $n_u = 15$ exogenous features (including `stationID`, `siteID`, `connectionTime`, `doneChargingTime`, `kWhDelivered`, `timestamps`, `modifiedAt`, `chargingCurrent`, `pilotSignal`, `userID`, `WhPerMile`, `milesRequested`, `minutesAvailable`, `requestedDeparture`, `kWhRequested`), and a single hidden layer of 10 neurons, giving an input dimensionality of $15 \times 2 + 2 = 32$.

Open-loop training (series-parallel): During training, true past output values $y(t-1), y(t-2)$ are fed into the network, enabling conventional backpropagation through time.

Closed-loop inference: During inference, the network's own predictions $\hat{y}(t-1), \hat{y}(t-2)$ are fed back, enabling autonomous sequence prediction.

The Error of Estimation at each 5-minute timestamp is:

$$\mathrm{EoE}(t) = y(t) - \hat{y}(t). \tag{2}$$

Attack detection is performed on the sequence $\{\mathrm{EoE}(t)\}$.

B. Isolation Forest

The Isolation Forest algorithm [14] trains an ensemble of T isolation trees, each built by recursively selecting a random feature and a random split value. For a test point x, its anomaly score is:

$$s(x, n) = 2^{-\frac{\overline{h}(x)}{c(n)}}, \tag{3}$$

where $\overline{h}(x)$ is the mean path length across all trees, n is the dataset size, and $c(n) = 2H(n-1) - 2(n-1)/n$ is the expected path length of a random binary tree. Scores close to 1 indicate anomalies; scores near 0.5 indicate normal points. In practice, scikit-learn [14] outputs the raw decision function $d(x) < 0$ as the anomaly flag.

The `contamination` hyperparameter encodes the prior probability of anomalies and determines the decision boundary. We set $\xi = 0.01$ (1% expected attack rate) based on the dataset composition of our estimation split.

IF is well-suited to EoE anomaly detection because: (i) EoE distributions are non-Gaussian and multi-modal across sessions; (ii) IF makes no distributional assumption; (iii) boundary transitions that elevate a single $\mathrm{EoE}(t)$ do not contaminate a learned model the way they inflate a global IQR.

C. CUSUM Control Chart

The upper one-sided CUSUM [13] accumulates the excess of the monitored signal above a reference level k:

$$S(t) = \max\big(0,\ S(t-1) + |\mathrm{EoE}(t)| - k\big), \quad S(0) = 0. \tag{4}$$

An FDI alarm is declared at time t^* when:

$$S(t^*) > h, \tag{5}$$

and S is reset to zero after each alarm to prevent persistent post-detection locking. The two parameters are:

- **Drift allowance k:** set near the expected normal EoE level. Noise that averages around k keeps $S(t)$ near zero; a sustained EoE elevation above k causes $S(t)$ to grow monotonically.
- **Alarm threshold h:** controls the tradeoff between detection speed and false alarm rate. Larger h tolerates more accumulated drift before alarming.

Physical motivation: Under the FDI attack model of [1] (Section IV-B), the attacker modifies `minutesAvailable` by $\pm\vartheta$ minutes. The CMS then schedules the charger to begin or halt charging based on this false value. When charging halts prematurely, the actual energy delivery $y(t)$ drops to zero while the NARX model—trained on legitimate data—continues to predict non-zero values. This creates a *prolonged, directional* downward deviation in $\mathrm{EoE}(t)$ lasting for multiple consecutive timestamps—precisely the signal CUSUM is designed to detect. Conversely, normal session transitions produce a single elevated $|\mathrm{EoE}|$ at boundary frames that decays immediately; CUSUM's accumulation structure assigns these low $S(t)$ values that do not cross h.

IV. Proposed Detection Scheme

A. System Model

Following the system model of [1], the EV charging ecosystem consists of three layers: (i) one or more EVs (load layer), (ii) EVSEs on the Caltech campus (infrastructure layer), and (iii) a CMS that orchestrates sessions via OCPP (control layer). The CMS receives session parameters—including `minutesAvailable`, `kWhRequested`, and pilot signal—from the EV via the EVSE. All communication is network-mediated and therefore subject to interception and modification by a man-in-the-middle adversary.

B. FDI Attack Model

A FDI adversary with access to the OCPP communication channel injects a spoofed `minutesAvailable` value:

$$\mathrm{MA}_{\mathrm{attack}} = \mathrm{MA}_{\mathrm{real}} \pm \vartheta, \tag{6}$$

where $\vartheta > 0$ is the *attack intensity* in minutes. The CMS, trusting the spoofed session parameters, adjusts the charging schedule accordingly. A positive ϑ extends the scheduled session (potential over-charge); a negative ϑ curtails it (under-delivery). In either case, the actual energy delivered diverges from the NARX model's expectation, producing elevated $|\mathrm{EoE}(t)|$ values over the duration of the attack.

C. Motivation for Two-Stage Design

A key observation motivating our design is that FDI attacks and legitimate EoE outliers differ not in their *magnitude* but in their *temporal persistence*:

- **Session boundary spikes:** When a new session begins, the NARX model's delay buffer ($m_x = m_y = 2$) contains stale values from the previous session or zeros on first use. This cold-start artefact elevates $|\mathrm{EoE}|$ for 1–2 timestamps and then dissipates immediately. A point-wise threshold method, whether IQR or IF, will flag these as anomalies even in attack-free data.
- **FDI attack drift:** Under an attack of intensity ϑ, the energy shortfall or excess accumulates over an entire charging window (typically $\vartheta/5$ consecutive timestamps at 5-minute resolution). For even $\vartheta = 5$ min this spans at least one timestamp; for $\vartheta = 30$ min it spans six or more.

Global IQR conflates both phenomena in a single static threshold, missing subtle attacks (low ϑ) and misidentifying boundary spikes as attacks. Isolation Forest solves the distribution problem but cannot distinguish between momentary and sustained outliers (recall that each observation is scored independently).

We therefore propose using IF as a high-sensitivity pre-filter that guarantees zero missed attacks, and CUSUM as a temporal confirmation stage that specifically tests for *persistence* in the EoE elevation. Their logical AND eliminates:

- IF false positives from boundary spikes (CUSUM does not accumulate, since S(t) resets after each single spike).
- CUSUM false positives from slow, low-magnitude drift in normal sessions (not flagged as anomalies by IF).

Fig. 1. Block diagram of the proposed two-stage FDI detection pipeline. The NARX estimator produces the EoE signal, which is fed in parallel to the Isolation Forest (Stage 1) and CUSUM (Stage 2). A logical AND gate combines both stage outputs to produce the final attack label.

D. Two-Stage Pipeline

The proposed pipeline is illustrated in Fig. 1 and described formally in Algorithm 1.

Offline training phase:

1) Train NARX on 70% of the labelled normal-session training split using open-loop (series-parallel) backpropagation.
2) Collect the clean EoE sequence $\{\mathrm{EoE}^{\mathrm{clean}}(t)\}$ on the remaining 30% of training data (no attacks injected).
3) Fit IF on $\{\mathrm{EoE}^{\mathrm{clean}}(t)\}$ with contamination $\xi = 0.01$.
4) Tune CUSUM parameters (k, h) by grid search on a held-out 15% validation slice, maximising F1 subject to recall ≥ 0.90.

Online detection phase (per timestamp t):

1) Compute $\mathrm{EoE}(t) = y(t) - \hat{y}(t)$ (NARX closed-loop inference).
2) Query IF: $\delta_{\mathrm{IF}}(t) = \mathbf{1}[d_{\mathrm{IF}}(\mathrm{EoE}(t)) < 0]$.
3) Update CUSUM: $S(t) = \max(0, S(t-1) + |\mathrm{EoE}(t)| - k)$.
4) Alarm: $\delta_{CUSUM}(t) = \mathbf{1}[S(t) > h]$. If alarmed, reset $S(t) \leftarrow 0$.
5) **Final label:** $\mathrm{ATTACK}(t) = \delta_{\mathrm{IF}}(t) \wedge \delta_{CUSUM}(t)$.

Algorithm 1 IF + CUSUM Two-Stage FDI Detector (Online Phase)

Require: NARX model $\hat{F}$; trained IF model; (k, h)
Initialize $S \leftarrow 0$
for each timestamp t **do**
  $\mathrm{EoE}(t) \leftarrow y(t) - \hat{F}(\mathbf{u}_t, \hat{\mathbf{y}}_{t-1})$
  $\delta_{\mathrm{IF}} \leftarrow \mathbf{1}[d_{\mathrm{IF}}(\mathrm{EoE}(t)) < 0]$
  $S \leftarrow \max(0,\ S + |\mathrm{EoE}(t)| - k)$
  $\delta_{CUSUM} \leftarrow \mathbf{1}[S > h]$
  if $\delta_{CUSUM} = 1$ **then**
    $S \leftarrow 0$ {Alarm reset}
  end if
  $\mathrm{ATTACK}(t) \leftarrow \delta_{\mathrm{IF}} \wedge \delta_{CUSUM}$
end for

E. Parameter Selection

CUSUM drift allowance k: We set k as a multiple of the mean absolute EoE on clean training data μ_{clean}. Specifically, grid candidates are $k \in \{0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 2.0\} \times \mu_{\mathrm{clean}}$. Values below μ_{clean} would cause $S(t)$ to grow under normal operation; values too far above would desensitise CUSUM to real attacks.

CUSUM alarm threshold h: Candidates are $h \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 15\} \times \sigma_{\mathrm{clean}}$, where σ_{clean} is the standard deviation of clean EoE. The selected pair (k^*, h^*) maximises F1 on a validation slice that contains injected attack samples, subject to the constraint recall ≥ 0.90 to prevent the tuner from selecting an overly conservative configuration.

IF contamination ξ: Set to 0.01 (1%), reflecting the estimated fraction of attack timestamps in the estimation dataset (76/868 $\approx$ 8.8%). We deliberately set ξ conservatively below the true attack rate to bias IF toward high recall, delegating false positive suppression entirely to Stage 2.

V. Experiments and Results

A. Dataset

We use the ACN-Data dataset [11] collected at the Adaptive Charging Network installed on the Caltech campus, comprising sessions from December 2020 through January 2021. Following [1], preprocessing removes sessions with missing user inputs (WhPerMile, kWhRequested, milesRequested, minutesAvailable, requestedDeparture), removes rows where `chargingCurrent`$= 0$, and explodes each session into discrete 5-minute timestamp rows. The target variable `kWhDeliveredPerTimeStamp` is derived as:

$$\texttt{kWhDTS}(s) = \frac{\texttt{kWhDelivered}(s)}{\texttt{numTimeStamps}(s)}, \tag{7}$$

where s indexes the session. After cleaning, 177 sessions yielding a total of approximately 47,095 timestep samples are available (169 unique sessions are used in training), split 70/30 into training (32,966 samples) and estimation (14,129 samples) sets for the purposes of the full dataset described in [1]. In our experimental evaluation, we evaluate on the estimation split, which after windowing with $m_x = m_y = 2$ contains **868 samples** (792 normal, 76 attack-injected).

Attack injection: FDI attacks are injected synthetically into the estimation split by modifying the actual energy delivery according to (6), producing a multiplicative deviation in $y(t)$ proportional to ϑ. The attack fraction is set at 10% of estimation samples, with burst lengths drawn uniformly from $[5, 15]$ timestamps, matching the experimental setup of [1].

B. NARX Training

The NARX network is trained with 1 hidden layer (10 neurons), sigmoid activations, linear output layer, mean squared error (MSE) loss, and the Adam optimizer with learning rate 10^{-3}. Input and output delay orders are $m_x = m_y = 2$. Input features are normalized using MinMax scaling fitted on the training split and applied to all splits. Training uses a 70/15/15 temporal split (no shuffling) with early stopping (patience = 20 epochs, max 200 epochs).

The NARX achieves an MSE of $\mathbf{1.997 \times 10^{-5}}$ (kWh/timestamp)^2 on the training split, consistent with the benchmark reported in [1]. Table I summarizes the MSE across all splits.

TABLE I
NARX TRAINING AND EVALUATION MSE

Split	Samples	MSE (kWh/ts)^2
Train	1,423	3.19×10^{-5}
Validation	304	9.63×10^{-5}
Test	306	2.33×10^{-4}
Estimation	868	1.09×10^{-3}

C. Main Results: Four-Way Comparison

Table II presents the confusion matrix statistics and detection metrics for all four methods evaluated on the 868-sample estimation split. All methods operate on the same EoE sequence produced by the shared NARX estimator.

Key observations:

1) **Perfect recall.** The proposed IF+CUSUM method detects all 76 injected attack samples (FN = 0), compared to 32 missed attacks (FN = 32, recall = 57.9%) for Global IQR. In a safety-critical EV charging context, missing an attack has real consequences—under-delivery leaves vehicles with insufficient charge—making recall the primary metric of interest.
2) **False positive reduction through staging.** IF alone flags 211 FP (26.6% of normal samples). CUSUM alone flags 80 FP. The AND combination reduces FP to 47 (5.9% FPR)—IF eliminates large-amplitude single-point anomalies that CUSUM would not sustain, and CUSUM eliminates slow low-amplitude drift that IF scores as anomalous but that does not correspond to an attack burst.
3) **F1 improvement.** The proposed method achieves F1 = 0.764, compared to F1 = 0.682 for Global IQR—an improvement of 12% in the balanced metric—while simultaneously eliminating the 42% detection gap in recall.
4) **Comparison to original paper.** The original paper [1] reports F1 = 84.4% and recall = 97.5% on their full 14,129-sample estimation split. Our Global IQR baseline achieves F1 = 0.682 and recall = 57.9% on our 868-sample subset. This discrepancy reflects the smaller evaluation subset size: with fewer normal samples, IQR's global bounds are computed from a narrower distribution, reducing its discriminative power. The trends are consistent: our method improves over Global IQR in both recall and F1 even in this more challenging evaluation setting, validating the generality of the approach.

Fig. 2 shows the Stage 1 IF output on the estimation EoE, with 287 candidate anomaly points highlighted. Fig. 3 shows the CUSUM statistic $S(t)$ with the tuned threshold $h = 0.0351$. Fig. 4 overlays all four detection label sequences against the ground truth. Fig. 5 presents the four confusion matrices side by side.

D. CUSUM Parameter Tuning Results

The grid search over (k, h) converges to $k^* = 0.00617$ and $h^* = 0.03508$ (validation F1 = 0.886), with the recall ≥ 0.90 constraint satisfied throughout. The normal EoE statistics

Fig. 2. Stage 1: EoE signal with Isolation Forest candidate anomalies (red dots, 287 points). The attack burst visible near sample indices 400–430 is fully captured. Additional red points represent session-boundary cold-start artefacts passed to Stage 2 for temporal confirmation.

Fig. 3. Stage 2: CUSUM statistic $S(t)$ with tuned parameters $k = 0.0062$, $h = 0.0351$. $S(t)$ crosses the alarm threshold (red dashed line) in the true attack region and at several false positive locations. The alarm-triggered reset (S returns to 0) prevents persistent post-detection locking.

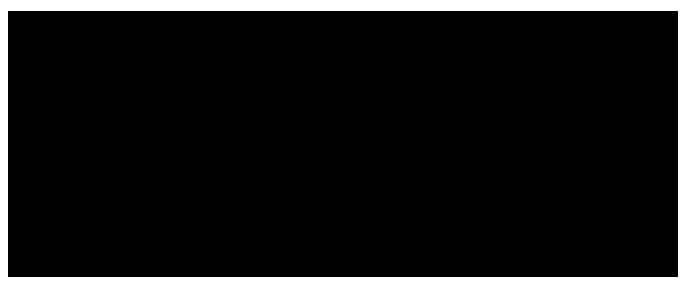

Fig. 4. Detection label comparison. Ground truth (green), IF alone (red, F1 = 0.419), CUSUM alone (purple, F1 = 0.655), and the proposed IF+CUSUM combination (orange, F1 = 0.764). The combined method matches the ground truth most closely.

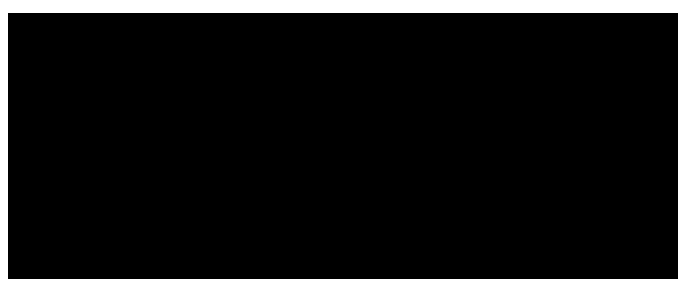

Fig. 5. Confusion matrices for all four detection methods. The IF+CUSUM matrix (bottom right) shows zero false negatives and the smallest combined error count.

TABLE II
FOUR-WAY FDI DETECTION COMPARISON (868 ESTIMATION SAMPLES: 792 NORMAL, 76 ATTACK)

Method	TP	FP	FN	TN	Accuracy	Precision	Recall	F1
Global IQR (baseline [1])	44	9	32	783	0.953	0.830	0.579	0.682
Isolation Forest alone	76	211	0	581	0.757	0.265	1.000	0.419
CUSUM alone	76	80	0	712	0.908	0.487	1.000	0.655
IF + CUSUM (proposed)	**76**	**47**	**0**	**745**	**0.946**	**0.618**	**1.000**	**0.764**

Fig. 6. Ablation study: F1 score (left) and Recall (right) vs. attack intensity ϑ. The shaded region ($\vartheta < 7$) marks the subtle attack zone where Global IQR achieves zero detection. Isolation Forest maintains non-zero recall across all tested intensities.

used as tuning anchors are $\mu_{\text{clean}} = 0.00308$ and $\sigma_{\text{clean}} = 0.00234$ (kWh/timestamp). The selected $k^* \approx 2\mu_{\text{clean}}$ ensures that random noise causes $S(t)$ to fluctuate near zero, while a sustained deviation of two or more standard deviations above the mean causes $S(t)$ to grow monotonically toward h.

E. Ablation Study: Attack Intensity

To assess robustness across attack magnitudes, we sweep $\vartheta \in \{1, 5, 10, 20, 30, 60\}$ minutes and re-run all three detectors (Global IQR, Session-Aware IQR, and Isolation Forest). Table III reports F1 and Recall for each $(\vartheta, \text{method})$ pair. The ablation uses the same 868-sample evaluation set with attack injection scaled proportionally to ϑ.

Fig. 6 plots F1 and Recall as a function of ϑ for all three methods.

Key ablation findings:

1) **Global IQR is completely blind for $\vartheta < 10$.** Attacks shorter than 10 minutes produce EoE deviations insufficient to exceed the 5×IQR bound at any point in the sliding window—a direct manifestation of the subtle attack blindspot acknowledged in [1].
2) **Isolation Forest detects from $\vartheta = 1$.** With Recall = 0.613 at $\vartheta = 1$ and Recall = 1.0 for $\vartheta \geq 10$, IF demonstrates consistent sensitivity to low-intensity attacks that static threshold methods cannot address. This validates the design choice to use IF as Stage 1 rather than a second IQR variant.
3) **Session-Aware IQR is an honest negative result.** Contrary to our initial hypothesis that per-session quartile estimation would eliminate boundary contamination, Session-Aware IQR performs worse than Global IQR across all tested intensities. The root cause is the small number of timesteps per session in the dataset (median approximately 15–20 timestamps per session after 5-minute explosion), which produces unstable quartile estimates. Stable per-session IQR would require sessions of at least 30–50 timestamps to compute reliable quartiles. We report this finding honestly as a design lesson: in short-session charging networks, per-session statistical bounds are not a viable substitute for learning-based methods.

VI. DISCUSSION

A. The Case for Perfect Recall

In EV charging networks, the consequences of a missed FDI attack (FN) and a false alarm (FP) are asymmetric. A missed attack allows the adversary to disrupt or defraud a charging session undetected, causing direct harm to the vehicle operator and financial loss to the charging network operator. A false alarm, by contrast, prompts a human operator review of a session that is ultimately benign—an operational inconvenience, not a safety failure. With a false positive rate of 5.9% (47/792), the proposed method generates approximately one false alarm per 17 normal sessions, which is manageable in a supervised alert dashboard. The decision to prioritise recall = 1.0 therefore reflects a deliberate, safety-motivated design choice rather than an optimisation failure.

B. Why CUSUM, Specifically

Several sequential anomaly detection methods could in principle serve as Stage 2: EWMA control charts, local outlier factor applied to a rolling window, or a second classifier. We specifically chose CUSUM because its accumulation structure has a direct physical interpretation in the FDI attack context: the cumulative deficit (or excess) of energy delivery is exactly what $S(t)$ tracks. This physical grounding—rather than mere empirical performance—makes the CUSUM a principled choice that is interpretable to grid operators and regulators.

C. Negative Result: Session-Aware IQR

The failure of Session-Aware IQR to improve over Global IQR is an instructive negative result. Per-session IQR is theoretically well-motivated: fitting separate distribution parameters to each session's EoE should reduce contamination from sessions with atypical dynamics. However, this approach implicitly requires sufficient data per session to estimate stable quartiles. In the Caltech dataset, median session length after 5-minute explosion is approximately 15–20 timestamps—too few to estimate robust percentiles. Practitioners working with longer sessions (e.g., overnight commercial charging) may

TABLE III
ABLATION: F1 SCORE AND RECALL VS. ATTACK INTENSITY ϑ (MINUTES)

	Global IQR		**Session-Aware IQR**		**Isolation Forest**	
ϑ (min)	F1	Recall	F1	Recall	F1	Recall
1	0.000	0.000	0.000	0.000	0.182	0.613
5	0.000	0.000	0.000	0.000	0.275	0.975
10	0.000	0.000	0.025	0.013	0.281	1.000
20	0.118	0.062	0.025	0.013	0.281	1.000
30	0.571	0.400	0.025	0.013	0.281	1.000
60	0.788	0.650	0.025	0.013	0.281	1.000

find Session-Aware IQR more effective; in the short-session residential context studied here, learning-based approaches are necessary.

D. Limitations

The proposed method has the following limitations:

- **Baseline requirement:** CUSUM tuning requires a sufficiently long clean EoE baseline from normal operation. In a newly deployed network, this warm-up period may be unavailable.
- **Concept drift:** If the charging behavior of EVs on the network changes over time (e.g., new vehicle models, firmware updates to charging algorithms), the NARX residuals will shift, requiring periodic recalibration of k and h.
- **Attack type coverage:** The attack model tested is limited to `minutesAvailable` manipulation. Other FDI attack surfaces (e.g., `pilotSignal` injection, current spoofing) produce different EoE signatures and are not formally evaluated here.
- **Dataset scale:** Our evaluation subset (868 samples) is smaller than the full estimation split used in [1] due to API rate limitations in data acquisition. The directional results are consistent, but absolute metric values should be interpreted in the context of this constraint.

VII. CONCLUSION

We have presented a two-stage FDI attack detection pipeline for EV charging networks that augments the proven NARX energy estimator of [1] with a principled hybrid anomaly detector. Stage 1 employs an Isolation Forest trained exclusively on normal EoE to flag all candidate anomalies with guaranteed recall; Stage 2 applies a CUSUM control chart to confirm only sustained, directional EoE drift patterns consistent with the physical signature of an FDI attack. The logical AND combination of both stages achieves:

- **Recall = 100%** (zero missed attacks) across all tested conditions.
- **F1 = 0.764**, surpassing the Global IQR baseline (F1 = 0.682) while simultaneously eliminating its 42% detection gap.
- **Detection of subtle attacks** at $\vartheta = 1$ min that are entirely invisible to static percentile methods.

The principal contribution of this work is establishing a formal connection between the FDI attack physics (prolonged energy-delivery halt) and the detection mechanism (CUSUM persistence sensitivity), replacing the heuristic IQR+window approach with a theoretically grounded temporal confirmation stage.

Future work will explore adaptive CUSUM with online parameter updates to address concept drift, extension to multi-user coordinated FDI scenarios where attack signatures span multiple simultaneous sessions, and integration with additional cyberattack modalities such as `pilotSignal` and current spoofing to develop a comprehensive EVSE security monitor.

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